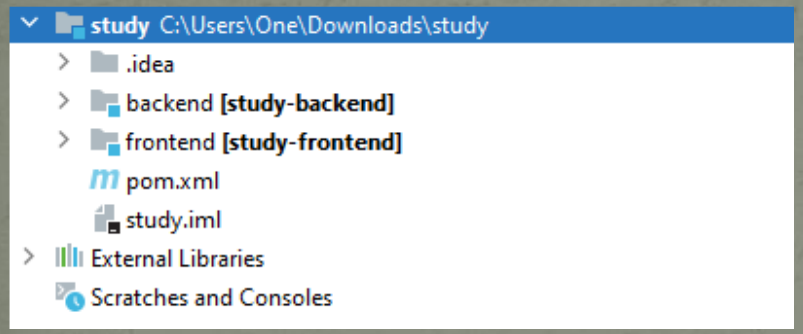
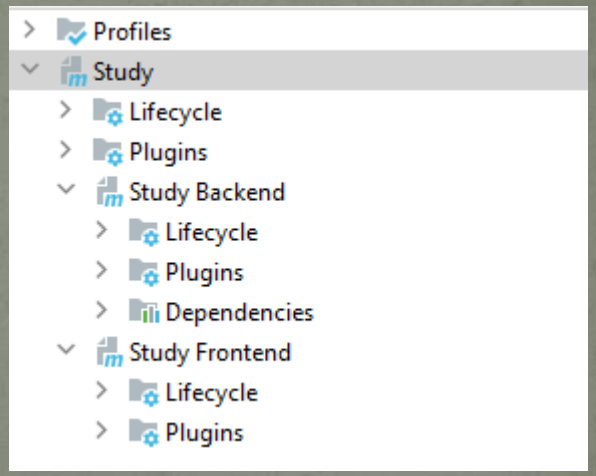
Разработка веб-приложений

Сборка и тестирование

Структура проекта

- При разработке SPA-приложений в простейшем случае при разделении функционала серверной и клиентской части проект состоит из двух модулей:
  - Front-end;
  - Back-end.





pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.0.2</version>
        <relativePath/> <!-- lookup parent from repository -->
    </parent>

    <groupId>com.ssau</groupId>
    <artifactId>study</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name>Study</name>
    <description>Study project for Spring Boot</description>
    <packaging>pom</packaging>

    <modules>
        <module>frontend</module>
        <module>backend</module>
    </modules>
</project>
```

pom.xml (frontend)

```
<project xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd"
        xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
   <modelVersion>4.0.0</modelVersion>
   <parent>
      <groupId>com.ssau</groupId>
      <artifactId>study</artifactId>
      <version>0.0.1-SNAPSHOT</version>
   </parent>

   <artifactId>study-frontend</artifactId>
   <version>0.0.1-SNAPSHOT</version>
   <name>Study Frontend</name>
   <description>Study project for Angular</description>
   <packaging>pom</packaging>
   ...
</project>
```

pom.xml (backend)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
            xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
   <modelVersion>4.0.0</modelVersion>
   <parent>
      <groupId>com.ssau</groupId>
      <artifactId>study</artifactId>
      <version>0.0.1-SNAPSHOT</version>
   </parent>

   <artifactId>study-backend</artifactId>
   <version>0.0.1-SNAPSHOT</version>
   <name>Study Backend</name>
   <description>Study project for Spring Boot</description>
   <properties>
      <java.version>17</java.version>
   </properties>
   ...
</project>
```

Плагины (frontend)

```
<plugin>
  <groupId>com.github.eirslett</groupId>
  <artifactId>frontend-maven-plugin</artifactId>
  <version>1.11.3</version>
  <configuration>
    <nodeVersion>v18.15.0</nodeVersion>
    <npmVersion>9.5.0</npmVersion>
    <workingDirectory>src/</workingDirectory>
  </configuration>
  <executions>
    <execution>
      <id>install node and npm</id>
      <goals>
        <goal>install-node-and-npm</goal>
      </goals>
    </execution>
    <execution>
      <id>npm install</id>
      <goals>
        <goal>npm</goal>
      </goals>
      <configuration>
        <arguments>install</arguments>
      </configuration>
    </execution>
...
```

Плагины (frontend)

```
...
    <execution>
      <id>npm run build</id>
      <goals>
        <goal>npm</goal>
      </goals>
      <configuration>
        <arguments>run build</arguments>
      </configuration>
    </execution>
    <execution>
      <id>prod</id>
      <goals>
        <goal>npm</goal>
      </goals>
      <configuration>
        <arguments>run-script build</arguments>
      </configuration>
      <phase>generate-resources</phase>
    </execution>
  </executions>
</plugin>
```

Плагины (backend)

```
<plugin>
   <artifactId>maven-resources-plugin</artifactId>
   <executions>
      <execution>
         <id>copy-resources</id>
         <phase>validate</phase>
         <goals>
            <goal>copy-resources</goal>
         </goals>
         <configuration>
            <outputDirectory>${project.build.directory}/classes/static/</outputDirectory>
            <resources>
               <resource>
                  <directory>${project.parent.basedir}/frontend/dist/frontend/</directory>
               </resource>
            </resources>
         </configuration>
      </execution>
   </executions>
</plugin>
```

Плагины (backend)

```
<plugin>
  <groupId>org.jacoco</groupId>
  <artifactId>jacoco-maven-plugin</artifactId>
  <version>0.8.7</version>
  <executions>
    <execution>
      <id>pre-unit-test</id>
      <goals>
        <goal>prepare-agent</goal>
      </goals>
      <configuration>
        <destFile>${project.build.directory}/jacoco.exec</destFile>
        <propertyName>surefireArgLine</propertyName>
      </configuration>
    </execution>
    <execution>
      <id>post-unit-test</id>
      <phase>test</phase>
      <goals>
        <goal>report</goal>
      </goals>
      <configuration>
        <dataFile>${project.build.directory}/jacoco.exec</dataFile>
        <outputDirectory>${project.build.directory}/jacoco</outputDirectory>
      </configuration>
    </execution>
    <execution>
      <id>pre-integration-test</id>
      <phase>pre-integration-test</phase>
      <goals>
        <goal>prepare-agent</goal>
      </goals>
      <configuration>
        <destFile>${project.build.directory}/jacoco-it.exec</destFile>
        <propertyName>failsafeArgLine</propertyName>
      </configuration>
    </execution>
    ...
```

Плагины (backend)

```
        ...
        <execution>
            <id>post-integration-test</id>
            <phase>post-integration-test</phase>
            <goals>
                <goal>report</goal>
            </goals>
            <configuration>
                <dataFile>${project.build.directory}/jacoco-it.exec</dataFile>
                <outputDirectory>${project.build.directory}/jacoco-it</outputDirectory>
            </configuration>
        </execution>
        <execution>
            <id>default-prepare-agent</id>
            <goals>
                <goal>prepare-agent</goal>
            </goals>
        </execution>
        <execution>
            <id>default-report</id>
            <phase>prepare-package</phase>
            <goals>
                <goal>report</goal>
            </goals>
        </execution>
    </executions>
</plugin>
```

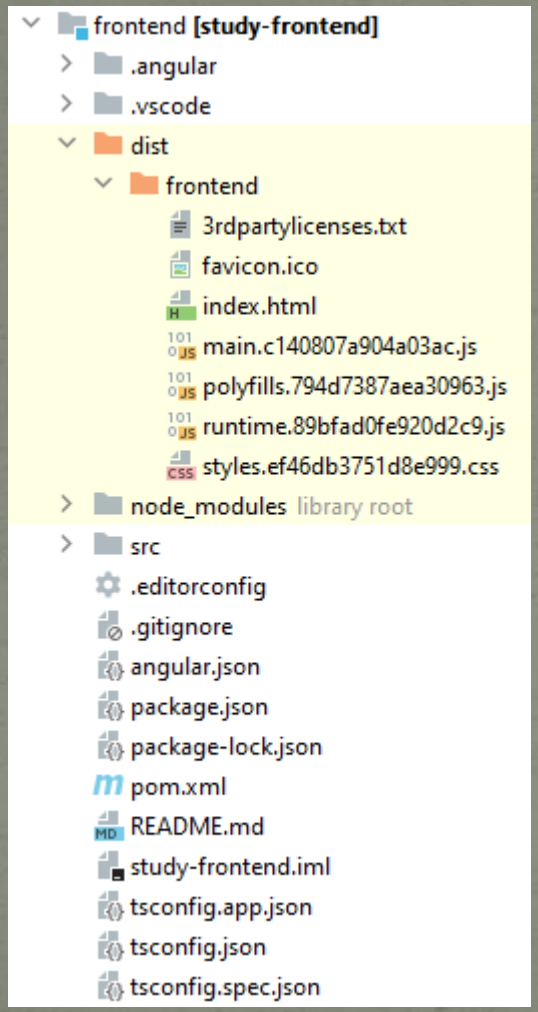
Плагины (backend)

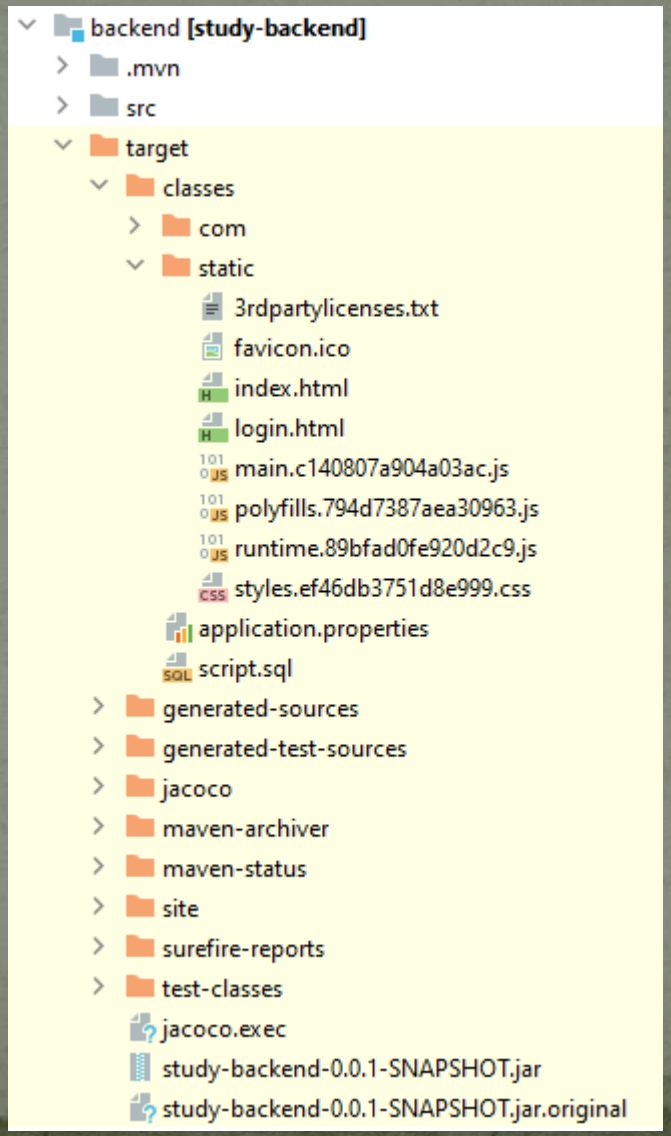
```
<plugin>
   <groupId>org.springframework.boot</groupId>
   <artifactId>spring-boot-maven-plugin</artifactId>
   <configuration>
      <excludes>
         <exclude>
            <groupId>org.projectlombok</groupId>
            <artifactId>lombok</artifactId>
         </exclude>
      </excludes>
   </configuration>
</plugin>
```

Сборка

```
[INFO] Scanning for projects...
[INFO] ------------------------------------------------------------------------
[INFO] Reactor Build Order:
[INFO]
[INFO] Study                                                      [pom]
[INFO] Study Frontend                                             [pom]
[INFO] Study Backend                                              [jar]
[INFO]
[INFO] --------------------------< com.ssau:study >--------------------------
[INFO] Building Study 0.0.1-SNAPSHOT                              [1/3]
[INFO] --------------------------------[ pom ]---------------------------------
...
[INFO] Study ........................................... SUCCESS [  0.109 s]
[INFO] Study Frontend ................................... SUCCESS [ 48.898 s]
[INFO] Study Backend .................................... SUCCESS [ 12.923 s]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:02 min
[INFO] Finished at: 2023-04-15T14:40:55+04:00
[INFO] ------------------------------------------------------------------------
```

Сборка


frontend [study-frontend]
.angular
.vscode
dist
frontend
3rdpartylicenses.txt
favicon.ico
index.html
main.c140807a904a03ac.js
polyfills.794d7387aea30963.js
runtime.89bfad0fe920d2c9.js
styles.ef46db3751d8e999.css
node_modules library root
src
.editorconfig
.gitignore
angular.json
package.json
package-lock.json
pom.xml
README.md
study-frontend.iml
tsconfig.app.json
tsconfig.json
tsconfig.spec.json


backend [study-backend]
.mvn
src
target
classes
com
static
3rdpartylicenses.txt
favicon.ico
index.html
login.html
main.c140807a904a03ac.js
polyfills.794d7387aea30963.js
runtime.89bfad0fe920d2c9.js
styles.ef46db3751d8e999.css
application.properties
script.sql
generated-sources
generated-test-sources
jacoco
maven-archiver
maven-status
site
surefire-reports
test-classes
jacoco.exec
study-backend-0.0.1-SNAPSHOT.jar
study-backend-0.0.1-SNAPSHOT.jar.original

Окружения

- Для добавления окружений в приложение используется команда:
  - ng generate environments
- По умолчанию создаются следующие файлы:
  - CREATE src/environments/environment.ts (31 bytes)
    CREATE src/environments/environment.development.ts (31 bytes)
    UPDATE angular.json (3007 bytes)
- В файле angular.json формируется настройка замены:
  - ```
    "configurations": {
      "development": {
        "fileReplacements": [ {
          "replace": "src/environments/environment.ts",
          "with": "src/environments/environment.development.ts"
        } ],
    ...
    ```
- Запуск клиента в режиме разработки для применения настроек development теперь необходимо сопровождать дополнительным параметром:
  - ng serve --configuration=development
    ```

Использование настроек

```
import { Component, OnInit } from '@angular/core';
import { HttpClient } from "@angular/common/http";

import { environment } from "../../environments/environment";

@Component({
  selector: 'app-students',
  templateUrl: './students.component.html',
  styleUrls: ['./students.component.css']
})
export class StudentsComponent implements OnInit {
  students: any[] = [];
  constructor(private http: HttpClient) {}

  ngOnInit(): void {
    this.http.get(environment.apiUrl + "/api/students").subscribe((data: any) => {
      this.students = data;
    });
  }
}
```

Интеграционные и модульные тесты

- Современные приложения достаточно сложны и содержат множество зависимостей. Интеграционное тестирование проверяет, что несколько компонентов системы работают вместе правильно. Оно сложно для автоматизации. Как правило, такие тесты требуют, чтобы вся или почти вся система была развернута и сконфигурирована на машине, на которой они выполняются.
- Модульное тестирование, или юнит-тестирование (unit testing) – процесс в программировании, позволяющий проверить на корректность отдельные модули исходного кода программы. Идея состоит в том, чтобы писать тесты для каждой нетривиальной функции или метода. Это позволяет достаточно быстро проверить, не привело ли очередное изменение кода к регрессии, то есть к появлению ошибок в уже оттестированных местах программы, а также облегчает обнаружение и устранение таких ошибок.

pom.xml

```
<dependency>
   <groupId>com.h2database</groupId>
   <artifactId>h2</artifactId>
   <scope>test</scope>
</dependency>
<dependency>
   <groupId>org.springframework.boot</groupId>
   <artifactId>spring-boot-starter-test</artifactId>
   <scope>test</scope>
</dependency>
<dependency>
   <groupId>junit</groupId>
   <artifactId>junit</artifactId>
   <scope>test</scope>
</dependency>
```

application.properties

```
spring.datasource.driver-class-name=org.h2.Driver
spring.datasource.url=jdbc:h2:mem:db;DB_CLOSE_ON
_EXIT=FALSE
spring.datasource.username=sa
spring.datasource.password=sa
```

Интеграционный тест

```
package com.ssau.study.controller;

import com.ssau.study.dto.GroupPojo;
import com.ssau.study.jpa.GroupRepository;
import com.ssau.study.orm.Group;
import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;

import java.util.List;

import static org.junit.jupiter.api.Assertions.assertEquals;

@SpringBootTest
class GroupControllerTest {
		@Autowired
		private GroupController groupController;

		@Autowired
		private GroupRepository groupRepository;
...
```

Интеграционный тест

```
…
        @BeforeEach
        public void before() {
                Group group = new Group();
                group.setId(1L);
                group.setName("Group 1");
                groupRepository.save(group);
                group = new Group();
                group.setId(2L);
                group.setName("Group 2");
                groupRepository.save(group);
        }

        @AfterEach
        public void after() {
                groupRepository.deleteAllInBatch();
        }

        @Test
        void testFindAllByName() {
                List<GroupPojo> groups = groupController.findAllByName("1");
                assertEquals(1, groups.size());
                assertEquals(1L, groups.get(0).getId());
                assertEquals("Group 1", groups.get(0).getName());
        }
}
```

Модульный тест

```
package com.ssau.study.service;

import com.ssau.study.dto.GroupPojo;
import com.ssau.study.jpa.GroupRepository;
import com.ssau.study.orm.Group;
import org.junit.jupiter.api.Test;

import java.util.ArrayList;
import java.util.List;

import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.mockito.Mockito.mock;
import static org.mockito.Mockito.when;

public class GroupServiceTest {
    @Test
    public void testFindAll() {
        GroupRepository groupRepository = mock(GroupRepository.class);
        Group group = new Group();
        group.setId(1L);
        group.setName("Group 1");
        group.setStudents(new ArrayList<>());
        when(groupRepository.findAll()).thenReturn(List.of(group));

        GroupService bookService = new GroupService(groupRepository, null);
        List<GroupPojo> groups = bookService.findAll(null);
        assertEquals(1, groups.size());
        assertEquals(1L, groups.get(0).getId());
        assertEquals("Group 1", groups.get(0).getName());
    }
}
```

После сборки

[INFO] Results:

[INFO]

[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

[INFO]

[INFO]

[INFO] --- jacoco-maven-plugin:0.8.7:report (post-unit-test) @ study-backend ---

[INFO] Loading execution data file C:\Users\One\Downloads\study\backend\target\jacoco.exec

[INFO] Analyzed bundle 'Study Backend' with 11 classes

[INFO]

[INFO] --- jacoco-maven-plugin:0.8.7:report (default-report) @ study-backend ---

[INFO] Loading execution data file C:\Users\One\Downloads\study\backend\target\jacoco.exec

[INFO] Analyzed bundle 'Study Backend' with 11 classes

[INFO]

[INFO] --- maven-jar-plugin:3.3.0:jar (default-jar) @ study-backend ---

[INFO]

[INFO] --- spring-boot-maven-plugin:3.0.2:repackage (repackage) @ study-backend ---

[INFO] Replacing main artifact with repackaged archive

[INFO]

[INFO] --- jacoco-maven-plugin:0.8.7:prepare-agent (pre-integration-test) @ study-backend ---

[INFO] failsafeArgLine set to -javaagent:C:\\Users\\One\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.7\\org.jacoco.agent-0.8.7-runtime.jar=destfile=C:\\Users\\One\\Downloads\\study\\backend\\target\\jacoco-it.exec

...

[[INFO] ----------

[INFO] BUILD SUCCESS

[INFO] ----------

Покрытие

Study Backend

Study Backend

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.ssau.study.dto		35 %		50 %	13	20	21	32	12	19	1	2
com.ssau.study.repository		19 %		n/a	4	5	10	12	4	5	0	1
com.ssau.study.orm		43 %		n/a	10	18	6	12	10	18	0	2
com.ssau.study.entity		0 %		n/a	9	9	6	6	9	9	1	1
com.ssau.study.controller		35 %		n/a	5	9	6	10	5	9	0	2
com.ssau.study.service		67 %		100 %	1	5	3	9	1	3	0	1
com.ssau.study		37 %		n/a	1	2	2	3	1	2	0	1
com.ssau.study.security		100 %		n/a	0	6	0	27	0	6	0	1
Total	256 of 522	50 %	1 of 6	83 %	43	74	54	111	42	71	2	11

Study Backend > com.ssau.study.service

com.ssau.study.service

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
GroupService		67 %		100 %	1	5	3	9	1	3	0	1
Total	19 of 59	67 %	0 of 4	100 %	1	5	3	9	1	3	0	1

Study Backend > com.ssau.study.service > GroupService

GroupService

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods
create(long, StudentPojo)		0 %		n/a	1	1	3	3	1	1
findAll(String)		100 %		100 %	0	3	0	5	0	1
GroupService(GroupRepository, StudentRepository)		100 %		n/a	0	1	0	1	0	1
Total	19 of 59	67 %	0 of 4	100 %	1	5	3	9	1	3

Покрытие

Study Backend > com.ssau.study.service > GroupService.java

GroupService.java

```
1.  package com.ssau.study.service;
2.
3.  import com.ssau.study.dto.GroupPojo;
4.  import com.ssau.study.dto.StudentPojo;
5.  import com.ssau.study.jpa.GroupRepository;
6.  import com.ssau.study.jpa.StudentRepository;
7.  import com.ssau.study.orm.Group;
8.  import com.ssau.study.orm.Student;
9.  import lombok.RequiredArgsConstructor;
10. import org.springframework.stereotype.Service;
11.
12. import java.util.ArrayList;
13. import java.util.List;
14.
15. @Service
16. @RequiredArgsConstructor
17. public class GroupService {
18.     private final GroupRepository groupRepository;
19.
20.     private final StudentRepository studentRepository;
21.
22.     public List<GroupPojo> findAll(String name) {
23.         List<GroupPojo> result = new ArrayList<>();
24.         for (Group group : name == null ? groupRepository.findAll() : groupRepository.findAllByNameContainingIgnoreCase(name)) {
25.             result.add(GroupPojo.fromEntity(group));
26.         }
27.         return result;
28.     }
29.
30.     public StudentPojo create(long groupId, StudentPojo pojo) {
31.         Student student = StudentPojo.toEntity(pojo);
32.         student.setGroup(groupRepository.findById(groupId).orElseThrow());
33.         return StudentPojo.fromEntity(studentRepository.save(student));
34.     }
35. }
```

Спасибо за внимание